

Question 1 (15 marks)

a) Express 1.46 as a fraction in its simplest form. 2

b) Express $\frac{1+\sqrt{8}}{1+3\sqrt{2}}$ with a rational denominator in its simplest form. 2

c) Simplify $\frac{x^2-x-6}{2x^3+16} \div \frac{x^2-3x}{2x+4}$ 3

d) Solve for x :

$$\frac{x+2}{2} = 4 - \frac{x-1}{3} \quad 2$$

e) Solve the following equations simultaneously: 3

$$y = x^3 + x^2 \quad \text{and} \quad y = x + 1$$

f) Solve

$$\frac{x}{2x-1} \geq 5 \quad 3$$

Question 2 (14 marks)

a) i) Simplify, expressing the answer with positive indices:

$$\frac{2^n \cdot 4^{n+1}}{8^{n-2}} \quad 2$$

ii) Solve for x :

$$\frac{3 \cdot 5^x - 1}{5^x + 2} = 2 \quad 2$$

iii) Find the value of x correct to 2 decimal places. 1

$$2^{-x} = 5$$

b) i) Evaluate $\log_4 3$, correct to 4 significant figures. 1

ii) Evaluate $\log_5 100 - \log_5 4$. 1

iii) Show that $3^{\log_3 9} = 9$ 1

c) Solve for x

$$\log_2(2x - 1) + \log_2(x - 4) = 2$$
 2

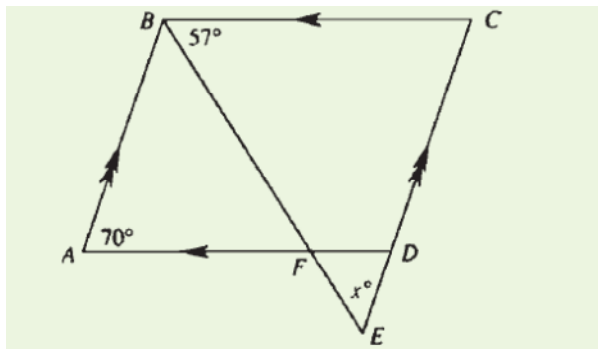
d) i) If $\log_{10} 2 = 0.3010$

Evaluate $\log_{10} (3.2)$ 2

ii) If $\log_a x = 1.8$ and $\log_a y = 2.4$, find $\log_a \left(\frac{a}{xy}\right)$ 2

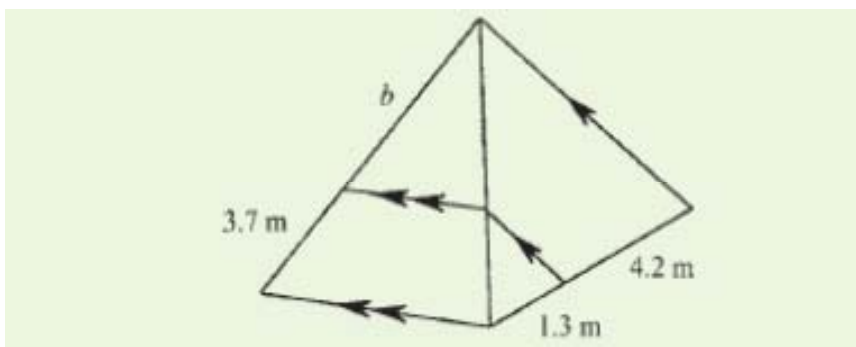
Question 3 (10 marks)

a) Copy the following diagram on to your answer sheet, and
Find the value of x , giving reasons for each step of your working out. 3



Not to scale

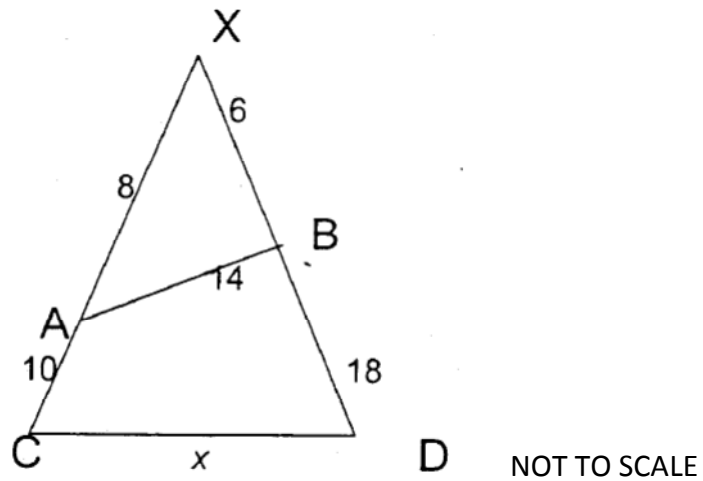
b) Copy the diagram on to your answer sheet and find the value of b .
(correct to 2 decimal places). 3



Not to scale

- c) Prove that $\triangle AXB$ is similar to $\triangle CXD$ and hence find the value of x .

4

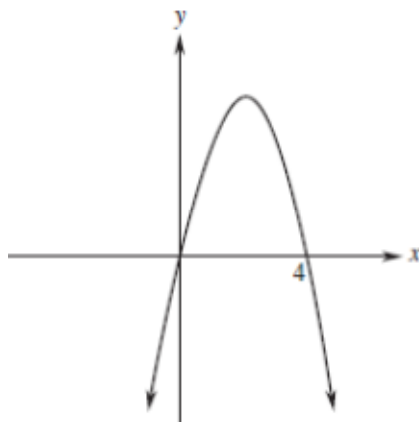


Question 4

(11 marks)

- a) i) State whether the graph below is odd, even or neither.

1



ii) If $f(x) = \begin{cases} 2x & , x \geq 0 \\ x^2 - 3 & , x < 0 \end{cases}$

find $f(5) - f(0) + f(-1)$

1

iii) State the domain and range of $y = \sqrt{2x - 4}$

2

iv) Show that $y = x^3 - x$ is an odd function.

1

b) i) On the same set of axes, sketch the graphs of $y = 2|x|$ and $y = |x + 3|$ 1

(note: your graph must be approximately 1/3 of a page in size and neatly presented)

ii) Find the point(s) of intersection of $y = 2|x|$ and $y = |x + 3|$ 2

c) Sketch the curve $y = 1 - \frac{4}{x^2 - 4}$ 3

END

Solutions - 2011 Task 1 - Ext 1 Preliminary

Comments

1 a) let $x = 1.4646 \dots$

(15)

$$100x = 146.46 \dots$$

$$99x = 145$$

$$x = \frac{145}{99} \quad \checkmark \quad \left(1 \frac{46}{99}\right)$$

b $\frac{(1 + \sqrt{8})}{(1 + 3\sqrt{2})} \times \frac{(1 - 3\sqrt{2})}{(1 - 3\sqrt{2})}$

$$= \frac{1 - 3\sqrt{2} + \sqrt{8} - 3\sqrt{16}}{1 - (9 \times 2)}$$

✓ ① x by the conjugate and with a rational denominator

$$= \frac{1 - 3\sqrt{2} + 2\sqrt{2} - 12}{-17}$$

$$= \frac{-11 - \sqrt{2}}{-17}$$

$$= \frac{11 + \sqrt{2}}{17}$$

✓ ① simplified answer

c) $\frac{x^2 - x - 6}{2x^3 + 16} \div \frac{x^2 - 3x}{2x + 4}$

$$= \frac{(x-3)(x+2)}{2(x^3+8)} \times \frac{2(x+2)}{x(x-3)} \quad \checkmark$$

① Partially factorised and simplified.

$$= \frac{(x+2)(x+2)}{x(x+2)(x^2-2x+4)} \quad \checkmark$$

① factorising sum of two cubes

$$= \frac{x+2}{x(x^2-2x+4)} \quad \checkmark = \frac{(x+2)}{x(x-2)^2}$$

① Answer

$$x(x^2 - 2x + 4)$$

①

$$d) \left[\frac{x+2}{2} = 4 - \frac{x-1}{3} \right] \times 6$$

$$3(x+2) = 24 - 2(x-1) \quad \checkmark$$

$$3x+6 = 24 - 2x+2$$

$$5x = 20$$

$$x = 4 \quad \checkmark$$

$$e) \begin{aligned} y &= x^3 + x^2 & \text{--- (1)} \\ y &= x+1 & \text{--- (2)} \end{aligned}$$

$$(2) = (1)$$

$$x^3 + x^2 = x + 1$$

$$x^2(x+1) - (x+1) = 0$$

$$(x+1)(x^2-1) = 0$$

$$(x+1)(x-1)(x+1) = 0$$

$$x = 1, -1 \quad \checkmark$$

when $x = 1$

$$y = 2 \quad \therefore (1, 2) \quad \checkmark$$

when $x = -1$

$$y = 0 \quad (-1, 0) \quad \checkmark$$

$$f) \quad \frac{x}{2x-1} \geq 5 \quad (\times \text{ b.s by } (2x-1)^2)$$

$$x(2x-1) \geq 5(2x-1)^2$$

$$5(2x-1)^2 - x(2x-1) \leq 0$$

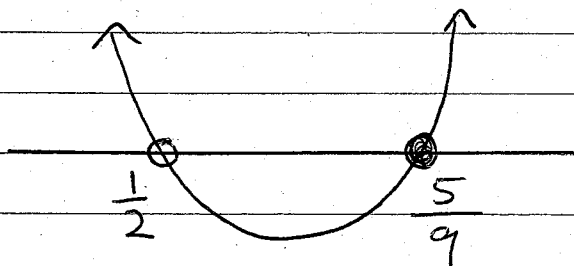
$$(2x-1)(5(2x-1) - x) \leq 0$$

$$(2x-1)(9x-5) \leq 0$$

✓ writing

$$(2x-1 \neq 0)$$

$$x \neq \frac{1}{2}$$



$$\frac{1}{2} < x \leq \frac{5}{9}$$

✓ values

✓ expression

① working

① values for x

① correct expression

$$2a) \quad \frac{2^n \cdot 4^n \cdot 4}{8^{n-2}}$$

(14)

$$i) \quad \frac{2^n \cdot 2^{2n} \cdot 2^2}{2^{3(n-2)}} \quad \checkmark$$

$$= \frac{2^n \cdot 2^{2n} \cdot 2^2}{2^{3(n-2)}} \quad \checkmark$$

$$= 2^{3n-3n} \cdot 2^{2+6}$$

$$= 2^8 \quad \checkmark \quad (256)$$

$$ii) \quad \frac{3 \cdot 5^x - 1}{(5^x + 2)} = 2 \quad \rightarrow$$

$$3 \cdot (5^x) - 1 = 2 \cdot (5^x) + 4 \quad \checkmark$$

$$5^x = 5^1$$

$$x = 1 \quad \checkmark$$

$$iii) \quad 2^{-x} = 5$$

$$-x = \frac{\ln 5}{\ln 2} \quad \checkmark$$

$$x = -2.32 \quad \checkmark$$

$$bi) \quad \log_4 3 = \frac{\ln 3}{\ln 4}$$

$$= 0.7925 \quad \checkmark$$

$$ii) \quad \log_5 100 - \log_5 4 = \log_5 \left(\frac{100}{4} \right)$$

$$= \log_5 25 = \log_5 5^2$$

$$= 2 \quad \checkmark$$

(4)

$$\begin{aligned}
 \text{ii)} \quad & \text{To show that } 3^{\log_3 9} = 9 \\
 & = 3^{\log_3 3^2} = 9 \\
 & = 3^{2 \log_3 3} = 3^2 \quad \checkmark
 \end{aligned}$$

① Answer with correct working.

$$c) \log_2 (2x-1) + \log_2 (x-4) = 2$$

$$(2x-1)(x-4) = 2^2 \quad \checkmark$$

$$2x^2 - 8x - x + 4 = 4$$

$$2x^2 - 9x = 0$$

$$x(2x-9) = 0$$

$$x \neq 0 \quad \therefore x = \frac{9}{2} \quad \checkmark$$

$$d i) \log_{10} 3.2 = \log_{10} \left(\frac{32}{10} \right)$$

$$= \log_{10} \left(\frac{2^5}{10} \right) \quad \checkmark$$

$$= 5 \cdot \log_{10} 2 - \log_{10} 10$$

$$= 5(0.3010) - 1$$

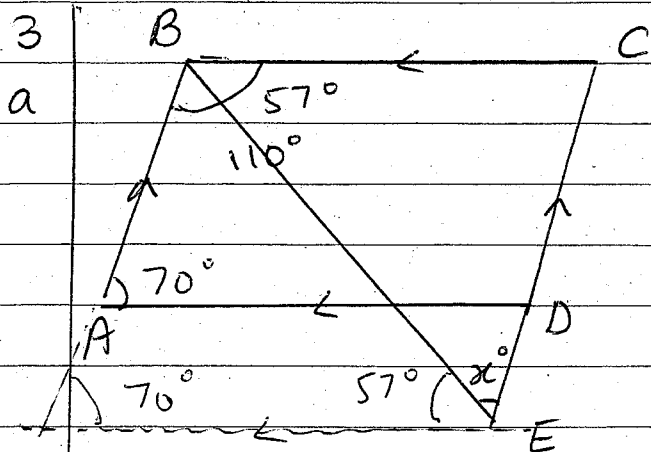
$$= 0.505 \quad \checkmark$$

$$\text{ii)} \log_a \left(\frac{a}{xy} \right) = \log_a a - [\log_a x + \log_a y] \quad \checkmark$$

$$= 1 - (1.8 + 2.4)$$

$$= -3.2 \quad \checkmark$$

(10)



$$\angle ABC = 110^\circ \text{ (co-interior to } \angle BAD)$$

✓

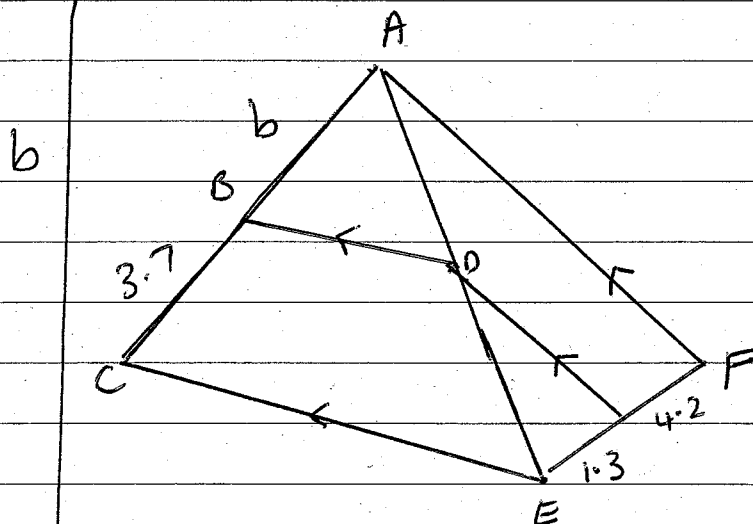
$$\begin{aligned} \angle ABE &= 110^\circ - 57^\circ \\ &= 53^\circ \end{aligned}$$

$$\therefore x^\circ = 53^\circ \text{ (alternate to } \angle ABE)$$

① working

① Answer

① Reason



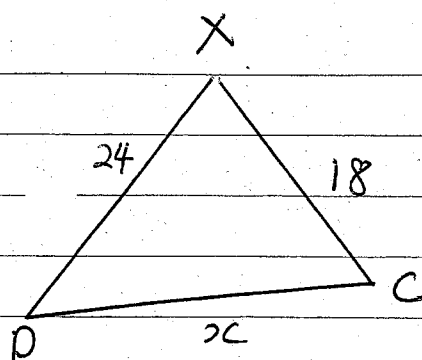
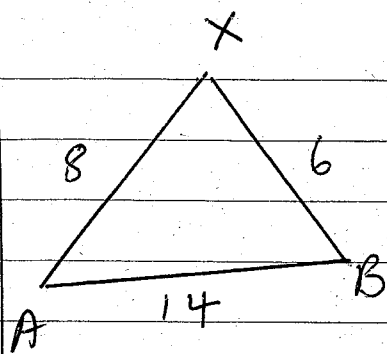
$$\frac{b}{3.7} = \frac{AD}{DE} = \frac{4.2}{1.3} \quad \checkmark$$

$$\frac{b}{3.7} = \frac{4.2}{1.3} \quad \checkmark$$

$$b = \frac{3.7 \times 4.2}{1.3} = 11.95 \text{ m} \quad \checkmark$$

(6)

c)



$\angle X$ is common

$$\frac{AX}{DX} = \frac{8}{24} = \frac{1}{3}$$

$$\frac{BX}{CX} = \frac{6}{18} = \frac{1}{3}$$

$\therefore \triangle AXB \parallel \triangle CXD$ (Corresponding sides have the same ratio.)

$$\therefore \frac{14}{x} = \frac{1}{3}$$

$$x = 3 \times 14 = 42$$

① Ratio

① Ratio

① Reason

① Answer

②

4 a

i) Neither ✓

$$ii) f(5) - f(0) + f(-1)$$

$$= 10 - 0 + 1 - 3$$

$$= 8$$

✓

⑦

$$\text{iii)} \quad y = \sqrt{2x-4}$$

$$2x-4 \geq 0$$

$$x \geq 2$$

$$D: x \geq 2 \quad \checkmark$$

$$R: y \geq 0 \quad \checkmark$$

$$\text{iv)} \quad y = x^3 - x$$

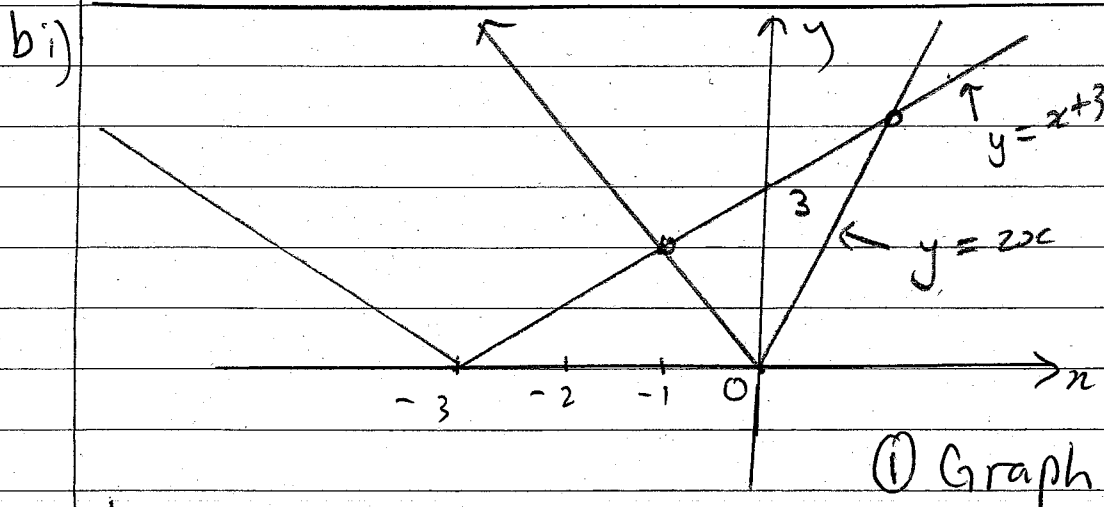
$$f(-x) = (-x)^3 - (-x)$$

$$= -x^3 + x$$

$$= -(x^3 - x) \quad \checkmark$$

$$= -f(x)$$

$\therefore f(-x) = -f(x)$, hence an odd function.



Intersection:

$$2x = x + 3$$

$$x = 3$$

$$y = 6$$

$$\therefore (3, 6)$$

$\checkmark$

$$-2x = (x + 3)$$

$$-3x = 3$$

$$x = -1$$

$$y = -1 + 3$$

$$= 2$$

$$\therefore (-1, 2) \quad \checkmark$$

(8)

$$c) \quad y = 1 - \frac{4}{x^2 - 4}$$

$$y = 1 - \frac{4}{(x-2)(x+2)}$$

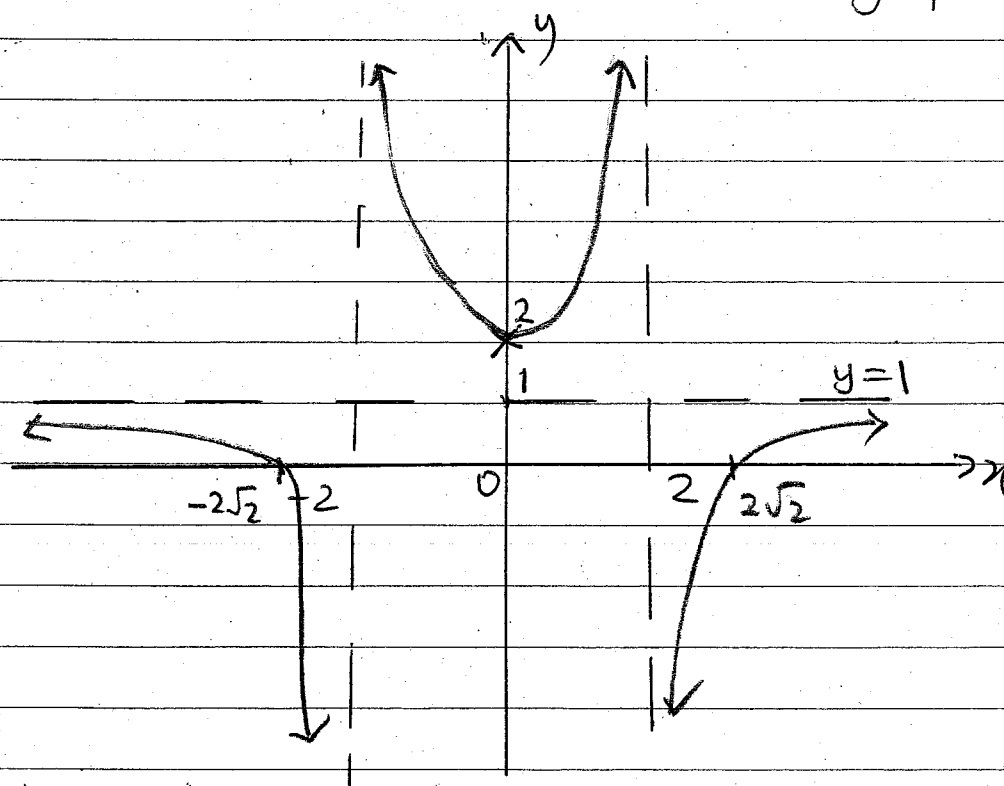
$$x \neq 2, -2$$

$\therefore$ Vertical asymptotes: $x=2, -2$

$$\frac{4}{(x-2)(x+2)} = 1 - y$$

$$4 = (1-y)(x-2)(x+2)$$

$\therefore y \neq 1$, hence $y=1$ is the horizontal asymptote



- ① Shape
- ① asymptotes
- ① intercepts

Intercepts:

$$x = 0$$

$$y = 2$$

$$y = 0$$

$$x^2 - 4 = 4$$

$$x^2 = 8$$

$$x = \pm 2\sqrt{2}$$